

Request for Comments: DECWAR Command Protocol Specification

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Category: Game Protocol

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Status: Draft Standard

Abstract

This document specifies the command protocol for DECWAR, a real-time multi-player space battle game. The protocol defines command syntax, input parsing rules, coordinate systems, and communication mechanisms for up to 10 concurrent players controlling starships in tactical combat scenarios.

1. Introduction

DECWAR is a real-time space battle simulation supporting 1-10 players across two primary factions (Federation and Empire) with optional neutral Romulan forces. This RFC documents the command interface protocol that enables players to control starships, manage resources, and engage in tactical operations within a 75x75 sector galaxy grid.

1.1 Design Goals

- Support real-time multiplayer interaction
- Provide consistent command abbreviation system
- Enable flexible coordinate specification methods
- Maintain game balance across different terminal speeds
- Support both novice and expert player interfaces

2. Command Protocol Overview

2.1 Input Processing Rules

All commands follow these parsing conventions:

- **Word Length:** Only first 5 characters of each word are stored
- **Separators:** Words separated by spaces, tabs, or commas
- **Line Termination:** `<CR>`, `<LF>`, `<VT>`, `<FF>`, `<ESC>`, or `^Z`

- **Echo Control:** `(^G)` toggles echo mode
- **Command Chaining:** Multiple commands separated by `(/)` (slash)
- **Comments:** Text after `(;)` treated as comments (except in TELL)
- **Repeat:** `(<ESC>)` as first character repeats previous command

2.2 Abbreviation System

- Ship names: Minimum 1 character
- Commands/keywords: Shortest unambiguous abbreviation (maximum 2 characters)
- Case insensitive processing

3. Coordinate System Specification

3.1 Coordinate Types

The protocol supports three coordinate specification methods:

3.1.1 Absolute Coordinates

```
[ABSOLUTE] <vertical_position> <horizontal_position>
```

- Default coordinate type
- Direct sector specification within 75x75 grid
- Example: `(37 45)` specifies sector at row 37, column 45

3.1.2 Relative Coordinates

```
RELATIVE <delta_vertical> <delta_horizontal>
```

- Offset from current ship position
- Positive values: up/right, negative values: down/left
- Example: `(RELATIVE 2 -3)` moves 2 sectors up, 3 sectors left

3.1.3 Computed Coordinates

```
COMPUTED <ship_name>
```

- Target ship's current location

- Requires operational computer system
- Restricted to terminals < 1200 baud
- Example: `COMPUTED BUZZARD` targets the Buzzard's position

3.2 Coordinate Output Formats

Configurable via SET OCDEF command:

- **ABSOLUTE:** Display as `vpos-hpos`
- **RELATIVE:** Display as `+/-dv,+/-dh`
- **BOTH:** Display both formats

4. Command Categories

4.1 Navigation Commands

4.1.1 MOVE Command

```
MOVE [coordinate_type] <vpos> <hpos>
```

- **Function:** Warp drive movement (factors 1-6)
- **Energy Cost:** Proportional to warp factor squared
- **Shield Penalty:** Double energy consumption if shields raised
- **Damage Risk:** Factors 5-6 risk engine damage
- **Status:** Changes condition to green

4.1.2 IMPULSE Command

```
IMPULSE [coordinate_type] <vpos> <hpos>
```

- **Function:** Emergency propulsion (warp factor 1)
- **Range:** Single sector movement only
- **Energy Cost:** Fixed low consumption
- **Status:** Changes condition to green

4.2 Combat Commands

4.2.1 PHASERS Command

PHASERS [coordinate_type] [energy] <vpos> <hpos>

- **Function:** Direct energy weapon
- **Range:** Maximum 10 sectors
- **Energy Cost:** 200 units (default), 50-500 units (specified)
- **Targeting:** Single target per command
- **Shield Interaction:** Additional 200 units if shields raised
- **Damage Probability:** 5% (200 units) to 65% (500 units)
- **Alert Status:** Triggers red alert

4.2.2 TORPEDOES Command

TORPEDOES [coordinate_type] <count> <v1> <h1> [<v2> <h2>] [<v3> <h3>]

- **Function:** Projectile weapon system
- **Count:** 1-3 torpedoes per burst
- **Range:** 8-10 sectors before self-destruct
- **Targeting:** Individual targeting per torpedo
- **Accuracy:** Affected by shield strength, computer damage, tube damage
- **Reload Time:** 2-4 seconds + (damage/100) per torpedo
- **Alert Status:** Triggers red alert

4.3 System Management Commands

4.3.1 SHIELDS Command

SHIELDS UP|DOWN
SHIELDS TRANSFER <energy>

- **Function:** Deflector shield control and energy transfer
- **Activation Cost:** 100 units ship energy
- **Transfer:** Bidirectional between ship and shield energy
- **Display Format:** (+n%) (up), (-n%) (down), (n%) (strength)

4.3.2 DOCK Command

DOCK [STATUS [device_names]]

- **Function:** Resupply at friendly base or planet
- **Requirements:** Adjacent to friendly installation
- **Resources:** Energy, torpedoes, repairs, damage reduction
- **Base Rates:** Ship energy +1000, Shield energy +500, Torpedoes +10
- **Planet Rates:** Half of base rates
- **Status:** Optional status report after docking

4.3.3 REPAIR Command

REPAIR [units]

- **Function:** Device damage restoration
- **Base Rate:** 30 units per turn + specified units
- **Docked Bonus:** 50% time reduction when docked
- **Default:** 50 units (100 if docked)
- **Scope:** All damaged devices simultaneously

4.4 Information Commands

4.4.1 SCAN Command

SCAN [direction] [range|vrange hrange] [WARNING]

- **Function:** Tactical sensor display
- **Default Range:** 10 sectors
- **Maximum Range:** 10 sectors
- **Directions:** UP, DOWN, RIGHT, LEFT, CORNER
- **Warning Mode:** Shows enemy weapon ranges with (!)
- **Symbols:** Federation (L,N,S,V,Y), Empire (C,D,H,J,W), Romulan (??)

4.4.2 STATUS Command

STATUS [attribute_list]

- **Function:** Ship status report
- **Attributes:** Condition, Location, Torpedoes, Energy, Damage, Shields, Radio
- **Conditions:** Green (normal), Yellow (low energy), Red (combat)

4.4.3 LIST Command

```
LIST [filter_keywords]
```

- **Function:** Comprehensive object information
- **Scope:** Ships, bases, planets
- **Filters:** Range, side, object type, specific names/locations
- **Default:** All available information

4.5 Communication Commands

4.5.1 TELL Command

```
TELL target_list;message
```

- **Function:** Sub-space radio communication
- **Targets:** Individual ships, teams (FEDERATION, EMPIRE), groups (ALL, FRIENDLY, ENEMY)
- **Requirements:** Operational sub-space radio
- **Restrictions:** Cannot be repeated with ESC key
- **Format:** Semicolon separates targets from message

4.5.2 RADIO Command

```
RADIO ON|OFF  
RADIO GAG|UNGAG <ship_name>
```

- **Function:** Communication system control
- **Global Control:** Enable/disable all communications
- **Selective Control:** Block/unblock specific ships
- **Status:** Affects incoming messages only

4.6 Strategic Commands

4.6.1 CAPTURE Command

CAPTURE [coordinate_type] <vpos> <hpos>

- **Function:** Planet acquisition
- **Time Cost:** 5 seconds + 1 second per enemy build level
- **Energy Cost:** 50 units per enemy build level
- **Requirements:** Adjacent to target planet

4.6.2 BUILD Command

BUILD [coordinate_type] <vpos> <hpos>

- **Function:** Planetary fortification construction
- **Time Cost:** 5-7 seconds
- **Build Limit:** 4 levels (5th converts to starbase)
- **Requirements:** Controlled planet
- **Benefits:** Increased damage resistance and firepower

4.7 Utility Commands

4.7.1 ENERGY Command

ENERGY <ship_name> <units>

- **Function:** Inter-ship energy transfer
- **Requirements:** Adjacent friendly ship
- **Efficiency:** 90% (10% transmission loss)
- **Automatic Limiting:** Respects recipient capacity

4.7.2 TRACTOR Command

TRACTOR <ship_name>
TRACTOR OFF

- **Function:** Ship towing capability
- **Requirements:** Adjacent friendly ship, both shields down

- **Energy Cost:** 3x normal movement consumption
- **Breaking Conditions:** Shields raised, torpedo hit, nova damage

5. Timing and Synchronization

5.1 Real-Time Constraints

Commands requiring real-time delays:

- **BUILD:** 5-7 seconds
- **CAPTURE:** 5 seconds + build penalties
- **DOCK:** 2-4 seconds
- **MOVEMENT:** 2-4 seconds
- **REPAIR:** $(0.08 * \text{repair_size})$ seconds

5.2 Weapon Cooldowns

5.2.1 Phaser Banks

- **Cooldown:** 3-6 seconds + $(\text{damage}/100)$
- **Banks:** Two independent banks
- **Firing Rate:** Once per cooldown period per bank

5.2.2 Torpedo Tubes

- **Reload Time:** 2-4 seconds + $(\text{damage}/100)$ per torpedo
- **Burst Limitation:** Complete reload required between bursts

5.3 Terminal Speed Compensation

Base timing adjusted for slowest terminal in game:

- **300 baud:** Maximum delays
- **1200+ baud:** Reduced delays
- **Computed Coordinates:** Available only to < 1200 baud terminals

6. Error Handling and Safety

6.1 Friendly Fire Prevention

Weapons officer automatically cancels:

- Phaser attacks on friendly targets
- Torpedo impacts on friendly installations
- Energy transfers exceeding capacity

6.2 Critical Condition Handling

6.2.1 Yellow Alert

- **Trigger:** Ship energy ≤ 1000 units
- **Effect:** Warning bell after each move
- **Status:** Automatic condition change

6.2.2 Red Alert

- **Triggers:** Combat engagement, weapons fire
- **Restrictions:** Cannot exit via ^C
- **Requirements:** Must use QUIT command

6.2.3 Life Support Failure

- **Condition:** Life support inoperative
- **Time Limit:** 5 stardates to dock or repair
- **Consequence:** Crew death if exceeded

7. Configuration and Customization

7.1 SET Command Options

```
SET OUTPUT LONG|MEDIUM|SHORT
SET SCAN LONG|SHORT
SET PROMPT NORMAL|INFORMATIVE
SET OCDEF ABSOLUTE|RELATIVE|BOTH
SET ICDEF ABSOLUTE|RELATIVE
SET NAME <player_name>
```

7.2 Persistent Settings

Settings may be stored in DECWAR.INI for automatic loading.

8. Security Considerations

8.1 Input Validation

- Command length restrictions prevent buffer overflow
- Coordinate bounds checking prevents invalid sector access
- Energy transfer limits prevent resource exploitation

8.2 Anti-Automation Measures

- Computed coordinates limited to slow terminals
- Real-time delays prevent scripted advantages
- ESC repeat restrictions on communication commands

9. IANA Considerations

This protocol operates within a closed game environment and requires no IANA registry assignments.

10. References

Normative References

[DECWAR] "DECWAR Version 2.2", December 25, 1979, Multi-Player Host System

Informative References

[HELP] DECWAR Online Help System, accessible via HELP command

[NEWS] DECWAR.NWS - Version enhancement documentation

[GRIPE] DECWAR.GRP - Bug report and suggestion system

Appendix A: Command Reference Quick Table

Command	Min Abbrev	Function	Real-Time
BASES	BA	List base information	No
BUILD	BU	Construct fortifications	Yes (5-7s)
CAPTURE	CA	Acquire planets	Yes (5s+)
DAMAGES	DA	Show device damage	No
DOCK	DO	Resupply at station	Yes (2-4s)
ENERGY	E	Transfer power	No
GRIPE	GR	Submit feedback	No
HELP	H	Display help	No
IMPULSE	I	Emergency movement	Yes (2-4s)
LIST	LI	Show object info	No
MOVE	M	Warp drive travel	Yes (2-4s)
NEWS	NE	Version information	No
PHASERS	PH	Energy weapons	Yes (cooldown)
PLANETS	PL	Planet information	No
POINTS	PO	Score breakdown	No
QUIT	Q	Exit game	No
RADIO	RA	Communication control	No
REPAIR	RE	Fix damage	Yes (variable)
SCAN	SC	Sensor display	No
SET	SE	Configure options	No
SHIELDS	SH	Shield control	No
SRSCAN	SR	Short range scan	No
STATUS	ST	Ship status	No
SUMMARY	SU	Object summary	No
TARGETS	TA	Enemy locations	No
TELL	TE	Send message	No
TIME	TI	Time information	No
TORPEDOES	TO	Projectile weapons	Yes (reload)
TRACTOR	TR	Towing beam	No
TYPE	TY	Show settings	No
USERS	U	Player list	No

Appendix B: Galaxy Symbols

Symbol	Meaning
L,N,S,V,Y	Federation ships
C,D,H,J,W	Empire ships
??	Romulan ship
<>	Federation starbase
) (	Empire starbase
@	Neutral planet
@F	Federation planet
@E	Empire planet
*	Star
●	Black hole
.	Empty sector
!	Danger zone (WARNING mode)

Authors' Address:

Based on original DECWAR implementation

Multi-Player Host System

December 25, 1979

Comments on this RFC may be submitted via the GRIPE command within DECWAR or directed to the game implementors.